Digital Interaction Framework

Trust Architecture for Self-Sovereign Digital Relationships

5	6	4	7	5
Layers	Stages	Policy Domains	ADRs	Autonomy Levels

TRUST ARCHITECTURE

Layered Trust Model

05 Interaction & Relationship
Per-exchange `ixn`
Transactional (ephemeral) vs Interactional (enduring). Shared context and activity binding within Enterprise Workflow Model.

04 Trust & Assurance
Progressive `ACDC`
Fidelity, confidence, and provenance. Risk calibration: transactional = lower threshold, interactional = progressive accumulation.

03 Technology
On rotation `icp/rot`
KERI infrastructure, WebAuthn passkeys, ACDC credentials, path history, encryption (AES-256-GCM/TLS 1.3), witness network.

02 Governance
Periodic `dip`
KEL governance (2-of-3 witness), credential governance, organisational rules, delegation framework for AI agents.

01 Policy Stacks
Infrequent `ADR`
Technical, Financial, Human, Agent policy domains. The most stable layer — foundation everything rests upon.

Layer Comparison

Layer	Actor	Primitive	Failure Mode
05 Interaction	Both parties	ixn	Scope creep
04 Trust	Verifier	ACDC	Confidence erosion
03 Technology	Infrastructure	icp, rot	Key compromise
02 Governance	Conductor	dip	Authority gap
01 Policy	Foundation	ADR	Non-compliance

RELATIONSHIP LIFECYCLE

Discovery to Dissolution

01 Discovering `icp` `oobi`
AID publication, capability matching, zero-trust entry. No trust assumed.

02 Co-Creating `ixn` `ACDC`
Bilateral credential exchange, context alignment, trust seeding through verification.

03 Proposing `dip` `ACDC`
Terms codification as ACDC schemas, governance agreement, agent delegation setup.

04 Using `ixn` `TEL`
Continuous verification, trust accumulation, anomaly detection and escalation.

05 Updating `rot` `ACDC`
Key rotation (pre-committed), credential refresh, scope renegotiation.

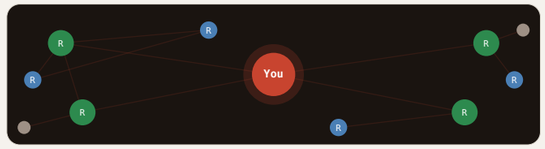
06 Archiving `TEL` `drt`
Credential revocation, agent deauthorisation, permanent audit trail preservation.

Trust Requirements by Stage

Stage	Trust	Credential Flow	Reversibility
Discovering	None	Outbound publish	Full
Co-Creating	Emerging	Bilateral exchange	High
Proposing	Threshold	Terms codify	Moderate
Using	Established	Continuous verify	Low
Updating	Maintained	Refresh/reissue	Moderate
Archiving	Residual	Revocation	Irreversible

AGENT NETWORK

Trust Tier Architecture



Primary AID (You) · Tier 1 — Direct · Tier 2 — Shared · Delegated

Tier 1 — Direct
Full ACDC chain, mutual KEL witness, direct signed comms, real-time TEL, agent delegation eligible.

Tier 2 — Shared
Group credential schema, issuer-mediated trust, attribute-based access, progressive elevation to Tier 1.

Teammate
Human Conductor + AI Agent. Root AID delegated dip AID. 70/30 energy split. Escalation returns control.

Dependable
Witness nodes (2-of-3), watchers (duplicity detection), OOBI discovery endpoints, TEL registries. Geo-distributed.

Protocol Infrastructure

KEL Append-only key event log. icp, rot, ixn, dip, drt

TEL Transaction Event Log — credential lifecycle tracking

OOBI Out-of-Band Introduction — endpoint discovery bootstrap

POLICY FRAMEWORK

Policy Stack Architecture

Technical
Ed25519, AES-256-GCM, TLS 1.3, HSM key storage, 2-of-3 witness,

Financial
Transaction limits, NPP/BPAY/SWIFT, AUSTRAC

Human
Conductor 70/30 model, graduated disclosure, OAIC

Agent
Action whitelists, 24h time limits, all ixn logged, anomaly escalation,

Architecture Decision Records

- `ADR-001` **KERI-First Identity** All identifiers are KERI AIDs. No alternative primary
- `ADR-002` **Passkey-Only AuthN** FIDO2/WebAuthn bound to KERI. No passwords, no
- `ADR-003` **ACDC Schema Gov.** Foundation-governed schemas. Graduated disclosure
- `ADR-004` **Witness 2-of-3** 3 witnesses, 2-of-3 threshold. Geo-distributed across 2+ AZs.
- `ADR-005` **Agent via dip** All agent authority via KERI dip with scoped time-limited
- `ADR-006` **Pre-Rotation** Next keys committed at inception. Not optional. Instant
- `ADR-007` **TEL for All** Every ACDC has TEL entry. Verifiers check before accepting.

Agent Autonomy Levels

Lvl	Name	Authority	Oversight
L1	Assistive	Read-only, suggest	Continuous
L2	Supervised	Execute w/ approval	Per-action
L3	Bounded	Within rules	Exception-based
L4	Autonomous	Self-directed	Periodic
L5	Strategic	Cross-domain	Outcome-based